6

Calibration

6.1 Introduction

6.1.1 History, Usage and Terminology

Calibration involves connecting one (or more) sets of variables together. Usually, one set (often called a *block*) is a series of physical measurements, such as some spectra or molecular descriptors, and the other contains one or more parameters such as the concentrations of a number of compounds or biological activity. Can we predict the concentration of a compound in a mixture spectrum or the properties of a material from its known structural parameters? Calibration provides the answer. In its simplest form, calibration is a form of regression, as discussed in Chapter 2, in the context of experimental design. There are several aims of calibration.

- The first aim is to simply form a functional model between two sets of measurements, for example, concentration and absorbance.
- The second aim is to find out whether this functional model is a good one, for example, by using linear model, can we adequately predict the concentration of an analyte from its absorbance?
- The third aim is to predict the properties of a new, unknown sample from making a series of measurements on it, for example, spectral absorbances.

Multivariate calibration has historically been a major cornerstone of chemometrics. However, there are a large number of diverse schools of thought, mainly dependent on people's background and the software they are familiar with. Many mainstream statistical packages do not contain the partial least squares (PLS) algorithm, whereas most specialist chemometric software is based around this method. PLS is one of the most publicised algorithms for multivariate calibration that has been widely advocated by many in chemometrics, following the influence of S. Wold, whose father first proposed this in the context of economics. A mystique surrounding PLS has developed, a technique with its own terminology, conferences and establishment. A number of commercial packages on the marketplace perform PLS calibration and result in a variety of diagnostic statistics. It is, though, important to understand that a major historic (and economic) driving force was near infrared spectroscopy (NIR), primarily in the food industry and in process analytical chemistry. Each type of spectroscopy and chromatography has its own features and problems; thus, much software was developed to tackle specific situations that may not necessarily be very applicable to other techniques such as, for example, chromatography, NMR or MS. In many statistical circles, NIR and chemometrics are almost inseparably intertwined. As time moves on, instruments improve in quality; thus, many of the computational approaches developed several decades ago to deal with problems such as background correction, and noise distributions, are not so relevant nowadays. However, for historical reasons, it is often difficult to distinguish between these specialist methods required to prepare data in order to obtain meaningful information from the chemometrics, and the actual calibration steps themselves. Despite this, chemometric approaches to calibration have very wide potential applicability throughout all areas of quantitative laboratory science, but NIR spectroscopists will definitely form one important readership base.

There are very many circumstances in which multivariate calibration methods are appropriate. The difficulty is that to develop a comprehensive set of data, analytical data collection for a particular situation may take a huge investment in resources and time; thus, the applications of multivariate calibration in some areas of science are much less well established than in others. It is important to separate the methodology that has built up around a small number of spectroscopic methods such as NIR, from the general principles. Probably several hundred favourite diagnostics are available to the professional user of PLS, for example, in NIR spectroscopy, yet each one has been

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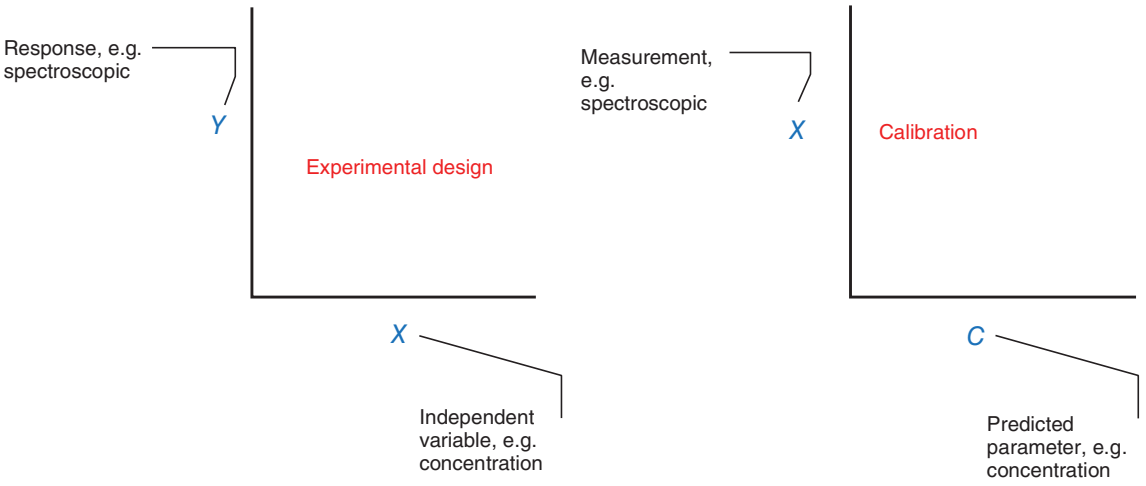


Figure 6.1 Different notations for calibration and experimental design as used in this book.

1. A series of 50 spectra are recorded at 120 wavelengths, and the concentrations of three compounds are independently known within these spectra.
   - **(a)** The *x* block has dimensions 50×120 and the *c* block 3×50.
   - **(b)** The *x* block has dimensions 50×120 and the *c* block 50×3.
   - **(c)** The *x* block has dimensions 120×50 and the *c* block 50×3.
   - **(d)** The *x* block has dimensions 120×50 and the *c* block 3×50.

6.1.2 Case Study

It is easiest to illustrate the methods in this chapter using a small case study, involving recording

- 25 EAS spectra at
- 27 wavelengths (220-350 nm at 5 nm intervals) and
- consisting of a mixture of 10 compounds (in our case, PAHs).

In reality, the spectra would usually be recorded at a higher digital resolution, but for illustrative purposes, we reduce the sampling rate. The aim is to predict the concentrations of individual PAHs from the mixture spectra. The spectroscopic data are presented in Table 6.1 and the corresponding concentrations of the compounds are presented in Table 6.2.

The methods in this chapter will be illustrated as applied to the spectroscopy of mixtures, as this is a common and highly successful application of calibration in chemistry. However, similar principles apply to a wide variety of calibration problems.

1. What are the dimensions of the *c* block in the case study of this section?
   - **(a)** 25×10
   - **(b)** 10×25
   - **(c)** 25×27
   - **(d)** 27×25

6.2 Univariate Calibration

Univariate calibration involves relating two single variables to each other and is often called *linear regression*. It is easy to perform using most data analysis packages.

Table 6.2 Concentrations of the 10 PAHs in the data in Table 6.1.

	Polyarene concentration (mg/l)									
Spectrum	Py	Ace	Anth	Acy	Chry	Benz	Fluora	Fluore	Nap	Phen
1	0.456	0.120	0.168	0.120	0.336	1.620	0.120	0.600	0.120	0.564
2	0.456	0.040	0.280	0.200	0.448	2.700	0.120	0.400	0.160	0.752
3	0.152	0.200	0.280	0.160	0.560	1.620	0.080	0.800	0.160	0.188
4	0.760	0.200	0.224	0.200	0.336	1.080	0.160	0.800	0.040	0.752
5	0.760	0.160	0.280	0.120	0.224	2.160	0.160	0.200	0.160	0.564
6	0.608	0.200	0.168	0.080	0.448	2.160	0.040	0.800	0.120	0.940
7	0.760	0.120	0.112	0.160	0.448	0.540	0.160	0.600	0.200	0.188
8	0.456	0.080	0.224	0.160	0.112	2.160	0.120	1.000	0.040	0.188
9	0.304	0.160	0.224	0.040	0.448	1.620	0.200	0.200	0.040	0.376
10	0.608	0.160	0.056	0.160	0.336	2.700	0.040	0.200	0.080	0.188
11	0.608	0.040	0.224	0.120	0.560	0.540	0.040	0.400	0.040	0.564
12	0.152	0.160	0.168	0.200	0.112	0.540	0.080	0.200	0.120	0.752
13	0.608	0.120	0.280	0.040	0.112	1.080	0.040	0.600	0.160	0.376
14	0.456	0.200	0.056	0.040	0.224	0.540	0.120	0.800	0.080	0.376
15	0.760	0.040	0.056	0.080	0.112	1.620	0.160	0.400	0.080	0.940
16	0.152	0.040	0.112	0.040	0.336	2.160	0.080	0.400	0.200	0.376
17	0.152	0.080	0.056	0.120	0.448	1.080	0.080	1.000	0.080	0.564
18	0.304	0.040	0.168	0.160	0.224	1.080	0.200	0.400	0.120	0.188
19	0.152	0.120	0.224	0.080	0.224	2.700	0.080	0.600	0.040	0.940
20	0.456	0.160	0.112	0.080	0.560	1.080	0.120	0.200	0.200	0.940
21	0.608	0.080	0.112	0.200	0.224	1.620	0.040	1.000	0.200	0.752
22	0.304	0.080	0.280	0.080	0.336	0.540	0.200	1.000	0.160	0.940
23	0.304	0.200	0.112	0.120	0.112	2.700	0.200	0.800	0.200	0.564
24	0.760	0.080	0.168	0.040	0.560	2.700	0.160	1.000	0.120	0.376
25	0.304	0.120	0.056	0.200	0.560	2.160	0.200	0.600	0.080	0.752

Py, pyrene; Ace, acenaphthene; Anth, anthracene; Acy, acenaphthylene; Chry, chrysene; Benz, benzanthracene; Fluora, fluoranthene; Fluore, fluorene; Nap, naphthalene; Phen, phenanthrene.

6.2.1 Classical Calibration

One of the simplest problems is to determine the concentration of a single compound using the response at a single detector, for example, a single spectroscopic wavelength or a chromatographic peak area.

Mathematically, a series of experiments can be performed to relate the concentration to spectroscopic measurements as follows:

$$\boldsymbol{x} \approx \boldsymbol{c}s$$

where, in the simplest case, $\boldsymbol{x}$ is a column vector consisting, for example, of absorbances at one wavelength for a number of samples and $\boldsymbol{c}$ is of the corresponding concentrations. Both vectors have I elements, equal to the number of samples. The scalar s relates these parameters and is determined by regression. Classically, most regression packages try to find s.

A simple method for solving this equation is to use the pseudo-inverse (see Section 2.2.3.3 for an introduction)

$$\boldsymbol{c}'\boldsymbol{x} \approx (\boldsymbol{c}'\boldsymbol{c})s$$

so

$$(\boldsymbol{c}'\boldsymbol{c})^{-1}\boldsymbol{c}'\boldsymbol{x} \approx (\boldsymbol{c}'\boldsymbol{c})^{-1}(\boldsymbol{c}'\boldsymbol{c})\, s$$

or

$$s \approx (\boldsymbol{c}'\boldsymbol{c})^{-1}\boldsymbol{c}'\boldsymbol{x} = \frac{\sum_{i=1}^{I} x_i c_i}{\sum_{i=1}^{I} c_i^2}$$

original graph (Figure 6.2) just to be sure, and this percentage appears reasonable. Providing a consistent measure is used throughout, all percentage errors will be comparable. Several other indicators of goodness-of-fit are also available, such as the squared Pearson's correlation coefficient between measured and predicted often denoted by R^2, but this is a big topic that we only summarise in this section.

This approach to calibration, while widely used throughout most branches of science, is nevertheless not always appropriate in all applications. We may want to answer the question 'can the absorbance in a spectrum be employed to determine the concentration of a compound?' It is not always the most appropriate approach to use an equation that predicts the absorbance from the concentration when our experimental aim is the reverse. In other areas of science, the functional aim might, for example, be to predict an enzymic activity from its concentration. In the latter case, classical calibration, as outlined in this section, results in the correct functional model. Nevertheless, most chemists employ classical calibration, providing the errors are approximately normal, and there are no significant outliers, all the different univariate methods should result in approximately similar conclusions.

For a new or unknown sample, however, the concentration could be estimated (approximately) by using the inverse of the slope or

$$\hat{c} = 3.32\, x$$

1. For a set of observations, $\sum_{i=1}^{I} x_i c_i = 2.58$ and $\sum_{i=1}^{I} c_i^2 = 3.98$. What is the predicted value of c for a value of x of 0.75 using classical calibration and no intercept term?
   (a) 1.16
   (b) 0.49

6.2.2 Inverse Calibration

Although classical calibration is widely used, it is not always the most appropriate approach, for two main reasons. First of all, the ultimate aim is usually to predict the concentration (or independent variable) from the spectrum or chromatogram (response) rather than *vice versa*. The second relates to error distributions. The errors in the response are often due to instrumental performance. Over the years, instruments have become more reproducible. The independent variable (often concentration) is usually determined via a procedure involving weighing, dilution and so on and is often by far the largest source of uncertainty (error). The quality of volumetric flasks, syringes and so on has not improved dramatically over the years, whereas the sensitivity and reproducibility of instruments has increased many fold. Classical calibration fits a model so that all errors are considered to be in the response (Figure 6.4(a)), whereas a more appropriate assumption is that errors are primarily in the measurement of concentration (Figure 6.4(b)).

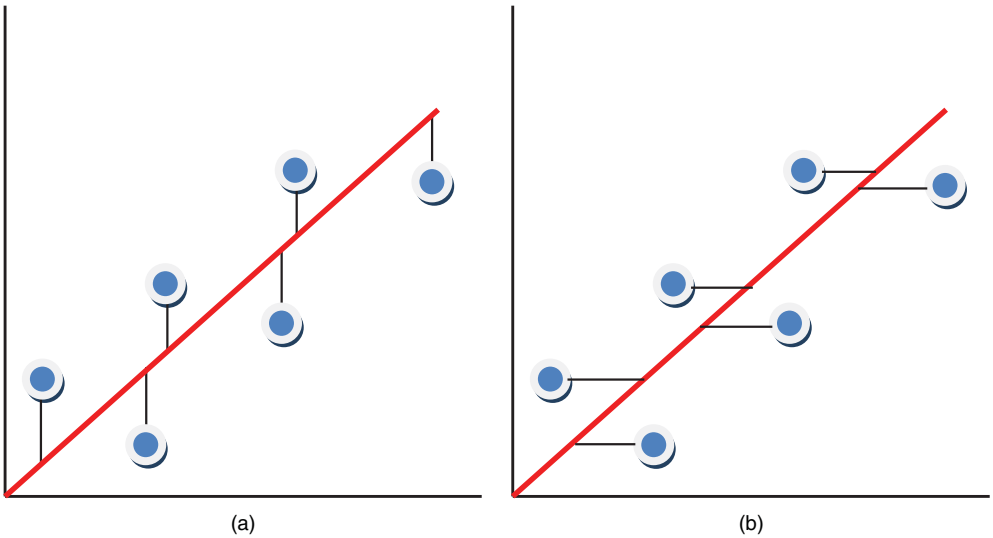


Figure 6.4 Difference between errors in (a) classical and (b) inverse calibration.

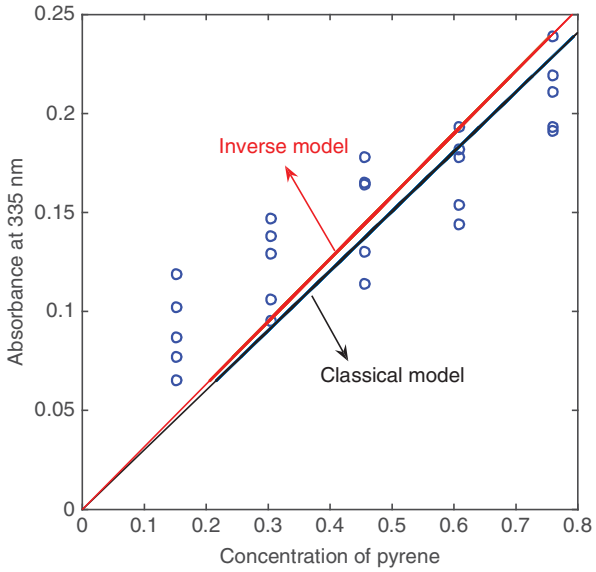


Figure 6.5 Best-fit straight lines for classical and inverse calibration, data for pyrene at 335 nm, no intercept, forcing the model through the origin.

1. For the example of question 1, Section 6.2.1, $\sum_{i=1}^{I} x_i^2 = 1.79$. What is the value of b in the relationship $\boldsymbol{c} \approx \boldsymbol{x}b$ for inverse calibration and no intercept term?
   (a) 1.44
   (b) 0.69

6.2.3 Intercept and Centring

In many situations, it is appropriate to include extra terms in the calibration model. Most commonly, an intercept (or baseline) term is included to give an inverse model of the form

$$c \approx b_0 + b_1 x$$

which can be expressed in matrix/vector notation by

$$\boldsymbol{c} \approx \boldsymbol{X}\boldsymbol{b}$$

for inverse calibration, where $\boldsymbol{c}$ is a column vector of concentrations and $\boldsymbol{b}$ is a column vector consisting of two numbers, the first equal to b_0 (the intercept) and the second equal to b_1 (the slope). $\boldsymbol{X}$ is now a matrix of two columns, the first of which is a column of 1's, the second the absorbances.

Exactly the same principles can be employed for calculating the coefficients, as discussed in Section 6.2.2, but in this case, $\boldsymbol{b}$ is a vector rather than scalar, and $\boldsymbol{X}$ is a matrix rather than a vector, so that using matrix notation,

$$\boldsymbol{b} \approx (\boldsymbol{X}'\boldsymbol{X})^{-1}\boldsymbol{X}'\boldsymbol{c}$$

or

$$\hat{c} = -0.17 + 4.23x$$

Note that the coefficient for b_1 is quite different to that for b, as calculated in Section 6.2.2. One reason is that there are still a number of interferents, from the other PAHs, in the spectrum at 335 nm, and these are modelled partly by the intercept term. The models of the previous sections force the best-fit straight line to pass through the origin. A better fit can be obtained if this condition is not required. The new best-fit straight line is presented in Figure 6.6 and results, visually, in a much better fit to the data.

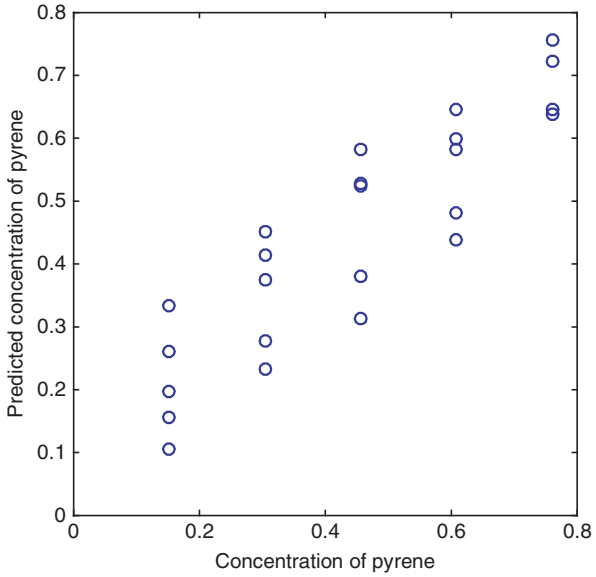


Figure 6.7 Predicted (vertical) versus known (horizontal) concentrations using methods of Section 6.2.3.

It is quite common to centre both sets of variables for this reason, the calculations being mathematically simpler than including an intercept term. Note that both blocks must be centred, and the predictions are of the concentrations minus their mean; hence, the mean concentration must be added back to return to the original values.

In calibration, it is common to plot a graph of predicted versus observed concentrations as presented in Figure 6.7. This looks superficially similar to that of the previous figure, but the vertical scale is different and the graph crosses the origin (providing the data have been mean centred). There is a variety of potential graphical output and it is important not to get confused between each type of graph, but to distinguish each type of information carefully.

It is important to realise that the predictions for the method described in this section differ to those obtained for the uncentred data. It is also useful to realise that similar methods can be applied to classical calibration, the details omitted for brevity, as it is recommended that inverse calibration is performed in normal circumstances in chemometrics.

1. Forty unique univariate measurements are recorded on a series of samples. A linear model including an intercept is used to fit the data.
   (a) There are 40 degrees of freedom for assessing the error.
   (b) There are 39 degrees of freedom for assessing the error.
   (c) There are 38 degrees of freedom for assessing the error.

6.3 Multiple Linear Regression

6.3.1 Multi-detector Advantage

Multiple linear regression (MLR) is an extension when more than one response is employed. There are three principal reasons for this. The first is that there may be more than one component in a mixture. Under such circumstances, it is usual to employ more than one response (the exception being if the concentrations of some of the components are correlated): for N components, at least N wavelengths should normally be used. The second is that each detector contains extra, and often complementary, information: some individual wavelengths in a spectrum may be influenced by noise or unknown interferants. Using, for example, a 100 wavelengths averages out the information and will often provide a better result than relying on a single wavelength. The third is that there may be an unknown number of factors that influence a response; thus, using very few variables will result in these factors being confounded and problems with finding unique and robust solutions.

When dealing with mixture spectra or chromatograms, if all the concentrations of all compounds are to be predicted, the number of variables or wavelengths must at least equal the number of compounds. However, if there are correlations either between the concentration profiles or the spectral profiles, it may not always be possible to resolve all the individual compounds. In our case study, the experiments are designed so that the concentrations are uncorrelated (using methods in Section 2.3.4) and the spectra all have unique characteristics.

1. A series of mixtures is known to contain 10 unique compounds and is studied spectroscopically.
   - **(a)** Exactly 10 wavelengths must be recorded and will always result in a unique solution for each compound.
   - **(b)** More than 10 wavelengths are always required for a unique solution.
   - **(c)** The number of wavelengths required to obtain a unique solution for all compounds depends on whether there is correlation between spectra and selective wavelengths.
   - **(d)** There may not be a unique solution for all the compounds in the mixture, no matter how many wavelengths, dependent on the experimental design and nature of the spectra.

6.3.2 Multi-wavelength Equations

In certain applications, equations can be developed that are used to predict the concentrations of compounds by monitoring at a finite number of wavelengths. A classical area is in pigment analysis by EAS, for example, in the area of chlorophyll chemistry. In order to determine the concentration of four pigments in a mixture, investigators recommend monitoring at four different wavelengths, and use an equation that links absorbance at each wavelength to concentration of the pigments.

In the PAH case study, only certain compounds absorb above 330 nm, the main ones being pyrene, fluoranthene, acenaphthalene and benzo[*a*]anthracene (note that the small absorbance due to a fifth component may be modelled as an interferant, although adding this to the model will, of course, result in better predictions but a more complicated model). It is possible to choose four wavelengths, preferably ones in which the absorbance ratios of these four compounds differ. The absorbance at wavelengths 330, 335, 340 and 345 nm of the pure compounds are illustrated in Figure 6.8. Of course, it is not necessary to select four sequential wavelengths, any four wavelengths would be sufficient, providing the four compounds are the main ones represented by these variables and their relative intensities differ, to give an $\boldsymbol{X}$ matrix with 25 rows and four columns.

Calibration equations can be obtained, as follows, using inverse methods.

- First, select the absorbances of the 25 spectra at these four wavelengths.
- Second, obtain the corresponding $\boldsymbol{C}$ matrix consisting of the relevant concentrations. These new (reduced) matrices are presented in Table 6.5.
- The aim is to find coefficients $\boldsymbol{B}$ relating $\boldsymbol{X}$ and $\boldsymbol{C}$ by $\boldsymbol{C} \approx \boldsymbol{X}\,\boldsymbol{B}$, where $\boldsymbol{B}$ is a 4×4 matrix, each *column* representing a compound and each *row* a wavelength. This equation can be solved using regression methods of Section 6.2.2, changing vectors and scalars to matrices, so that $\boldsymbol{B} = (\boldsymbol{X}'\boldsymbol{X})^{-1}\boldsymbol{X}'\boldsymbol{C}$ giving the matrix in Table 6.6.
- If desired, represent in equation form, for example, the first column of $\boldsymbol{B}$ suggests that

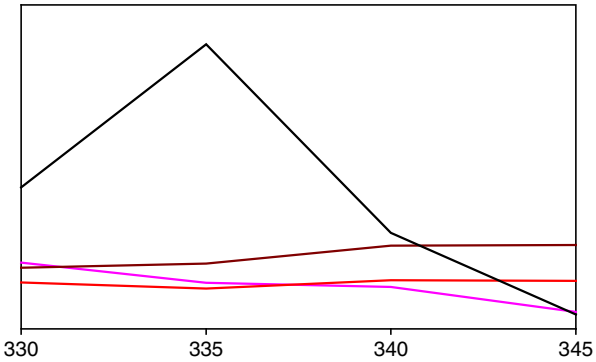


Figure 6.8 Absorbances of Pyr, Fluor, Benz and Ace between 330 and 345 nm.

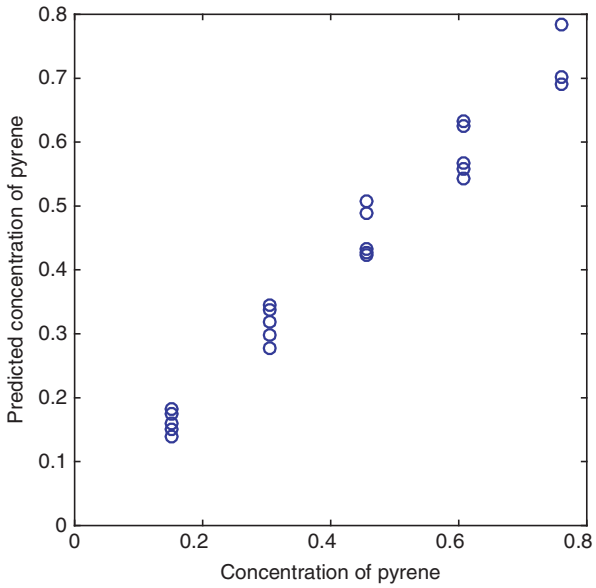


Figure 6.9 Predicted versus known concentration of pyrene, using a four-component model and the wavelengths 330, 335, 340 and 345 nm (uncentred).

However, as in univariate calibration, the coefficients obtained using both approaches may not be exactly equivalent, as each method makes different assumptions about the error structure. However, if the models from both methods differ substantially, this could be an indication of outliers, additional significant components and so on.

Note that the number of wavelengths must be more than or equal to the number of compounds in the mixture (or factors as appropriate). Mathematically, this is because the matrix $\boldsymbol{SS}'$ must be invertible. In addition, the concentrations and spectra of compounds must not be correlated, again chemical sense and mathematical sense come to the same conclusion – when there are complex multi-component mixtures, careful thought must be given to the design.

Equations where the number of wavelengths equals the number of analytes make assumptions that the concentrations of the significant analytes are all known and work well only if this is true. Applying to mixtures where there are significant unknown interferants can result in serious estimation errors.

1. A mixture consists of three components, recorded at three wavelengths and four samples. The $\boldsymbol{X}$ matrix is $\begin{bmatrix} 0.571 & 0.960 & 0.763 \\ 1.090 & 1.230 & 1.048 \\ 0.863 & 1.230 & 0.445 \\ 0.635 & 1.019 & 0.437 \end{bmatrix}$, where the columns represent the three wavelengths and the $\boldsymbol{C}$ matrix is $\begin{bmatrix} 0.78 & 0.35 & 1.04 \\ 0.52 & 0.90 & 1.76 \\ 0.39 & 1.21 & 0.50 \\ 0.44 & 0.88 & 0.23 \end{bmatrix}$, where the columns represent the three compounds.
   (a) The concentration of compound 1 can be approximated by
   $$-1.43A_1 + 1.01A_2 + 0.80A_3$$
   (b) The concentration of compound 1 can be approximated by
   $$-1.43A_1 + 1.40A_2 + 1.10A_3$$

6.3.3 Multivariate Approaches

The methods in Section 6.3.2 could be extended to all 10 PAHs, and with appropriate choice of 10 wavelengths may give reasonable estimates of concentrations. However, all the recorded wavelengths contain some information and there is no reason to restrict to only 10 selected wavelengths.

2. So long as the number of compounds of interest is always less than or equal to the number of wavelengths monitored, it is always possible to form an MLR model to predict the concentration well of these compounds in a series of mixtures?
   (a) True
   (b) False

6.4 Principal Components Regression

MLR-based methods have the disadvantage that all significant components (or compounds in our case) in the region of interest must be known. Principal components analysis (PCA)-based methods do not require details about the spectra or concentrations of all the compounds in a mixture, although it is important to make a sensible estimate how many significant components characterise a mixture, but not necessarily their chemical characteristics.

Principal components are primarily abstract mathematical entities and further details are described in Chapter 4. In multivariate calibration, the aim is to convert these to compound concentrations. Principal components regression (PCR) uses regression (sometimes also called *transformation* or *rotation*) to convert PC scores to concentrations. This process is often loosely called *factor analysis*, although terminology differs according to author and discipline. Note that although the chosen example in this chapter involves calibrating concentrations to spectral absorbances, it is equally possible, for example, to calibrate the property of a material to its structural features, or the activity of a drug to molecular parameters.

6.4.1 Regression

If the vector $\hat{\boldsymbol{c}}_n$ contains the estimated concentrations of compound n in a series of spectra (25 in this instance), it can be obtained from the PC scores matrix, $\boldsymbol{T}$, as follows:

$$\hat{\boldsymbol{c}}_n = \boldsymbol{T}\boldsymbol{r}_n$$

where $\boldsymbol{r}_n$ is a column vector whose length equals the number of PCs retained in the model, sometimes called a *rotation* or *transformation vector*. Ideally, the length of $\boldsymbol{r}_n$ should be equal to the number of compounds in the mixture (=10 in this case). However, noise, spectral similarities and correlations between concentrations and spectra, sometimes, make it hard to provide an exact estimate of the number of significant components: this topic has been discussed in Section 4.3.2. We will assume, for the purpose of this section, that 10 PCs are retained in the model.

The scores of the first 10 PCs are presented in Table 6.10, using raw data. Naturally, data could be pre-processed, including centring if desired.

The transformation vector can be obtained by using the pseudo-inverse of $\boldsymbol{T}$.

$$\boldsymbol{r}_n = (\boldsymbol{T}'\boldsymbol{T})^{-1}\boldsymbol{T}'\boldsymbol{c}_n$$

where $\boldsymbol{c}_n$ are the concentrations of compound n. Note that the matrix $(\boldsymbol{T}'\boldsymbol{T})$ is actually a diagonal matrix, whose elements consist 10 eigenvalues of the PCs and so each element of $\boldsymbol{r}$ could also be expressed as a summation

$$r_{na} = \frac{\sum_{i=1}^{I} t_{ia}c_{in}}{\sum_{a=1}^{A} g_a}$$

or even as a product of vectors

$$r_{na} = \frac{\boldsymbol{t}_a'\boldsymbol{c}_n}{\sum_{a=1}^{A} g_a}$$

We remind the reader of the main notation:

are necessary to model the data using methods such as cross-validation of percentage error, but after that, not every compound in the mixture needs to be modelled individually.

For a new sample, not necessarily part of the training set, we can predict the concentration of all analytes in the model as follows, first by estimating the scores using the loadings matrix, and then transforming these estimated scores

$$\hat{\boldsymbol{t}} = \boldsymbol{x}\boldsymbol{P}' \text{ hence } \hat{\boldsymbol{c}} = \hat{\boldsymbol{t}}\boldsymbol{r}$$

or in, one step, $\hat{\boldsymbol{c}} = \boldsymbol{x}\boldsymbol{B}$ where $\boldsymbol{B} = \boldsymbol{P}'\boldsymbol{r}$. Note that the matrix $\boldsymbol{B}$, which is used for estimation, differs according to the various methods described in this chapter.

1. A set of 30 mixture spectra recorded at 50 wavelengths consisting of eight main compounds and four PCs are retained for PCR. For each individual compound, a vector $\boldsymbol{r}$ is calculated. Its dimensions are
   (a) 8×1
   (b) 1×8
   (c) 4×1
   (d) 1×4

6.4.2 Quality of Prediction

A key issue in calibration is to determine how well the data have been modelled. In this section, we will discuss the training set or calibration error; Section 6.6 provides a further overview of how the approaches below can be used. We can now look at how the model changes for PCR as we change the number of principal components to help us decide how many PCs are suitable for the model.

Most look at how well the concentration is predicted, or the c (or according to some authors y) block of data.

The simplest method is to determine the sum of square of residuals between the true and predicted concentrations

$$S_c = \sum_{i=1}^{I} \left(c_{in} - \hat{c}_{in}\right)^2$$

where

$$\hat{c}_{in} = \sum_{a=1}^{A} t_{ia} r_{an}$$

for compound n using a principal components. The larger this error, the worse the prediction, hence the error reduces as more components are calculated.

Often, the error is reported as a RMSE, divided by the number of degrees of freedom (where A equals the number of PCs in the model),

$$RMSE = \sqrt{\frac{\sum_{i=1}^{I} (c_{in} - \hat{c}_{in})^2}{I - A}}$$

If the data are centred, a further degree of freedom is lost; hence, the sum of square residual is divided by $I{-}A{-}1$.

This error can also be presented as a percentage error,

$$RMSE_{\%} = 100E/\bar{c}_n$$

where $\bar{c}_n$ is the mean concentration in the original units. Sometimes, the percentage of the standard deviation is calculated instead, but in this section, we will compute errors as a percentage of the mean unless specifically stated.

It is also possible to report errors in terms of quality of modelling of spectra (or chromatograms), often called the x block error.

The quality of modelling of the spectra using PCA (the x variance) can likewise be calculated as follows:

$$S_x = \sum_{i=1}^{I} \sum_{j=1}^{J} \left(x_{ij} - \hat{x}_{ij}\right)^2$$

where

$$\hat{x}_{ij} = \sum_{a=1}^{A} t_{ia} p_{aj}$$

percentage variance remaining (or explained by each PC) or eigenvalues. Note a very small rise in the RMSE *c* error at four components: this is because the number of degrees of freedom changes more rapidly than the absolute error.

It is most common to use the *c* block error as this relates to how well the concentrations rather than the spectra are modelled, but sometimes comparing both gives a good idea of how many PCs are best to use in the model.

1. If a centred data set consists of 30 samples, 20 variables and is modelled by five PCs, how many degrees of freedom are there to assess the error in the predicted values of ***c***?
   (a) 30
   (b) 25
   (c) 24
2. A data set consists of 100 samples and 30 variables. A four PC model gives eigenvalues of 103, 70, 16 and 4 and the total sum of squares of the data is 215, there is no pre-processing. What is the $RMSE_x$?
   (a) 0.073
   (b) 0.271
   (c) 22
   (d) 4.69

6.5 Partial Least Squares Regression

PLS is often presented as the main regression technique used when there is multivariate data. In fact, its use is not always justified, and the originators of the method were well aware of this, but that being said, in some applications, PLS has been spectacularly successful. In some areas such as QSAR, or even metabolomics and psychometrics, PLS is an invaluable tool because the underlying factors have little or no physical meaning; hence, a linearly additive model in which each underlying factor can be interpreted chemically is not anticipated. In spectroscopy or chromatography, we usually expect linear additivity, and under such circumstances, simpler methods such as MLR are often adequate, providing there is fairly full knowledge of the system. However, PLS is always an important tool when there is partial knowledge of the data, a typical example being the measurement of protein in wheat by NIR spectroscopy. A model can be obtained from a series of wheat samples, and PLS will use typical features in this data set to establish a relationship to the known amount of protein. The basis is that if a training set is well designed, it will encompass the sort of samples that are likely to be encountered in the future. PLS models can be very robust, providing future samples do indeed contain similar features to the original data. Another example is the determination of vitamin C in orange juices using spectroscopy: a very reliable PLS model could be obtained using orange juices from a particular region of Spain, but what if some Brazilian orange juice is included? There is no guarantee that the model will perform well on the new data, as there may be different spectral features; hence, it is always important to be aware of the limitations of the method, particularly to remember that the use of PLS cannot compensate for poorly designed experiments or inadequate experimental data. Of course, it is possible to look for outliers that do not have the same correlation structure as the calibration or training set, but this is another layer of data analysis that many neglect.

An important feature of PLS is that it takes into account errors both in the concentration estimates and in the spectra. A method such as PCR assumes that the concentration estimates are error free. Much traditional statistics rests on this assumption that all errors are in the variables (spectra). If in medicine it is decided to determine the concentration of a compound in the urine of patients as a function of age, it is assumed that age can be estimated exactly, the statistical variation being in the concentration of a compound and the nature of the urine sample. Yet, in chemistry, there are often significant errors in sample preparation, for example, accuracy of weighings and dilutions and so the independent variable in itself also contains errors. Classical and inverse calibration force the user to choose which variable contains the error, whereas PLS assumes that it is equally distributed in both the *x* and *c* blocks and differs in this key assumption from the methods described above.

6.5.1 PLS1

The most widespread approach is often called *PLS1*. Although there are several algorithms, the main ones due to Wold and Martens, the overall principles are quite straightforward.

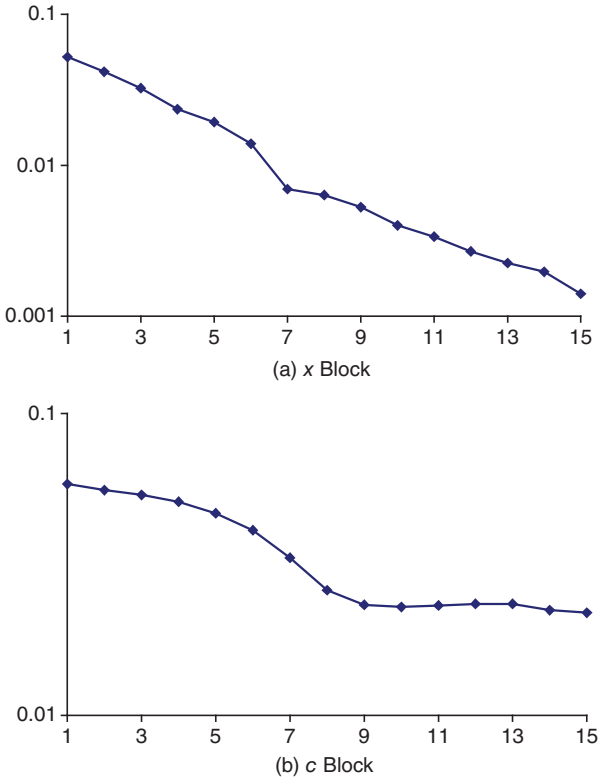


Figure 6.14 Residual errors in *x* and *c* blocks, PLS1 centred and acenaphthene.

contain errors in themselves, and what has been shown is that PLS forces the solution to model the *c* block better; it does not necessarily imply that the other methods are worse at discovering the truth. If, however, we have a lot of confidence in the experimental procedure to determine *c* (e.g. weighing and dilution), PLS will result in a more faithful reconstruction.

1. PLS1 loadings, like PC loadings, are always normalised and orthogonal.
   (a) True
   (b) False
2. Using a two-component model on a mean centred training set, the estimated PLS scores for a sample for compound A are [1.27 3.51] and the values of *q* for the first two components are 0.31 and 0.12. The mean concentration is 0.56 ml. What is its estimated concentration in the sample?
   (a) 1.37 ml
   (b) 0.25 ml
   (c) 0.81 ml

6.5.2 PLS2

An extension to PLS1 was proposed in the mid-1980s, called *PLS2*. In fact, there is little conceptual difference with PLS1, except that the latter uses a concentration matrix, $\boldsymbol{C}$, rather than concentration vectors for each individual compound in a mixture, and the algorithm is iterative. The equations above alter slightly in that $\boldsymbol{Q}$ is now a matrix of dimensions $A \times N$ and not a vector, so that

$$\boldsymbol{X} = \boldsymbol{TP} + \boldsymbol{E}$$

$$\boldsymbol{C} = \boldsymbol{TQ} + \boldsymbol{F}$$

The number of columns in $\boldsymbol{C}$ and $\boldsymbol{Q}$ are equal to the number of compounds (in our case) of interest. In PLS1, one compound is modelled at a time, whereas in PLS2, all known compounds can be included in the model simultaneously.

6.5.3 Multi-way PLS

Two-way data such as HPLC–DAD, LC–MS, LC–NMR and fluorescence excitation–emission spectroscopy are increasingly common in chemistry. Conventionally, either a univariate parameter (e.g. a peak area at a given wavelength) (methods in Section 6.2) or a chromatographic elution profile at a single wavelength (methods in Sections 6.3-6.5.2) is used for calibration, allowing the use of the regression techniques described above. However, additional information has been recorded for each sample, often involving both an elution profile and a spectrum. For example, a series of two-way chromatograms may be available and can be organised into a three-way array often visualised as a box, sometimes denoted by $\underline{\boldsymbol{X}}$ where the line underneath the array name indicates a third dimension. Each level of the box consists of a single chromatogram. Sometimes, these three-way arrays are called *tensors*, but tensors often have special properties in physics, which are unnecessarily complex and confusing to the chemometrician. We will use the notation of tensors only where it helps understand the existing methods.

Enhancements of the standard methods for multivariate calibration are required. Although it is possible to use methods such as three-way MLR, most chemometricians have concentrated on developing approaches based on PLS, which will be the focus below. Theoreticians have extended these methods to cases where there are several dimensions in both the x and c blocks, but the most complex practical case in chemistry is where there are three dimensions in the x block, as happens for a series of coupled chromatograms or in fluorescence excitation–emission spectroscopy, for example. A simple simulated numerical example is presented in Table 6.17, in which the x block consists of 4 two-way chromatograms, each of dimensions 5×6. There are three components in the mixture, the c block consisting of a 4×3 matrix. We will restrict the discussion for the case where each column of c is to be estimated independently (analogous to PLS1) rather than all in one go. Note that although PLS is by far the most popular approach for multi-way calibration, it is possible to envisage methods analogous to MLR or PCR, but they are rarely used.

6.5.3.1 Unfolding

One of the simplest methods is to create a single, long data matrix from the original three-way tensor. In the case of Table 6.17, we have four samples, which could be arranged as a $4 \times 5 \times 6$ tensor (or 'box'). The three dimensions will be denoted as I, J and K. It is possible to change the shape so that any binary combination of variables is converted to a new variable, for example, the intensity of the variable at $J = 2$ and $K = 3$ and the data can now be represented by $5 \times 6 = 30$ new variables and is the unfolded form of the original data matrix. This operation is illustrated in Figure 6.16.

It is now a simple task to perform PLS (or indeed any other multivariate approach), as discussed above. The 30 variables are centred and the predictions of the concentrations performed while increasing number of components are used (note that 3 is the maximum permitted for column centred data in this case, so this example is somewhat simple). All the methods described above can be applied.

An important aspect of three-way calibration involves centring, which can be rather complex. There are four different ways in which the data can be treated.

- No centring.
- Centre the columns in each $J \times K$ plane and then unfold with no further centring; hence, for example, $x_{1,1,1}$ becomes $390 - (390 + 635 + 300 + 65 + 835)/5 = -55$.
- Unfold the raw data and centre afterwards; hence, for example, $x_{1,1,1}$ becomes $390 - (390 + 488 + 186 + 205)/4 = 72.75$.
- Combine methods 2 and 3, start with centring as in step 2, then unfold and re-centre a second time.

These four methods are illustrated in Table 6.18 for the case of the $x_{i,1,1}$, the variables in the top left-hand corner of each of the 4 two-way data sets. Note that methods 3 and 4 yield radically different answers; for example, sample 2 has the highest value (=170.75) using method 3, but the lowest using method 4 (=−87.85).

Standardisation is also sometimes employed, but must be done before unfolding for meaningful results, an example might be in the GC–MS of a series of samples, each mass being of different absolute intensity. A sensible three-step strategy might be as follows.

- Standardise each mass in each individual chromatogram to provide I standardised matrices of dimensions $J \times K$
- Unfold
- Then, centre each of the variables.

Standardising at the wrong stage of the analysis can result in meaningless data; hence, it is always essential to carefully think of the physical (and numerical) consequences of any pre-processing which is far more complex and has far more options than for simple two-way data.

After this pre-processing, all the normal multivariate calibration methods can be employed.

1. A data set consists of 20 samples, recorded using excitation emission fluorescence consisting of 40 excitation and 50 emission wavelengths. The unfolded data set has dimensions
   (a) 20×40
   (b) 20×50
   (c) 20×200
   (d) 20×2000

6.5.3.2 Tri-linear PLS1

Some of the most interesting theoretical developments in chemometrics over the past few years have been in so-called *multi-way* or *multi-mode* data analysis. Many such methods have been available for some years, especially in the area of psychometrics, but a few do have relevance to chemistry and allied laboratory science. We will restrict the discussion in this chapter to tri-linear PLS1, involving a three-way x block and a single c variable. If there are several known calibrants, the simplest approach is to perform tri-linear PLS1 individually on each variable.

As centring can be quite complex for three-way data, and there is no inherent reason to do this, for simplicity, so for in the example of this section, we use uncentred data for illustrative purposes. The data in Table 6.17 can be considered to be arranged in the form of a cube, with three dimensions, I for the number of samples and J and K for the measurements.

Tri-linear PLS1 attempts to model both the x and c blocks simultaneously. In this review, we will illustrate the use with the algorithm of Section A.2.4, based on the methods proposed by de Jong and Bro.

Superficially, tri-linear PLS1 has many of the same objectives to normal PLS1, and the method as applied to the x block is often represented diagrammatically as in Figure 6.17, replacing 'squares' or matrices by 'boxes' or tensors, and replacing, where necessary, the dot product ('·') by something called a *tensor product* ('⊗'). The c block decomposition can be represented as per PLS1 and is omitted from the diagram for brevity. In fact, as we shall see, this is an oversimplification and is not an entirely accurate description of the method.

In tri-linear PLS1, for each component, it is possible to determine

- a scores vector ($\boldsymbol{t}$), of length I or 4 in this example,
- a weight vector, which has analogy to a loadings vector (${}^{j}\boldsymbol{p}$) of length J or 5 in this example, referring to one of the dimensions (e.g. time), whose sum of squares equals 1,
- another weight vector, which also has analogy to a loadings vector (${}^{k}\boldsymbol{p}$) of length K or 6 in this example, referring to the other one of the dimensions (e.g. wavelength) whose sum of squares also equals 1.

Superficially, these vectors are related to scores and loadings in normal PLS, but in practice, they are somewhat different concepts, a key reason being that more of these vectors are not orthogonal in tri-linear PLS1. In this chapter,

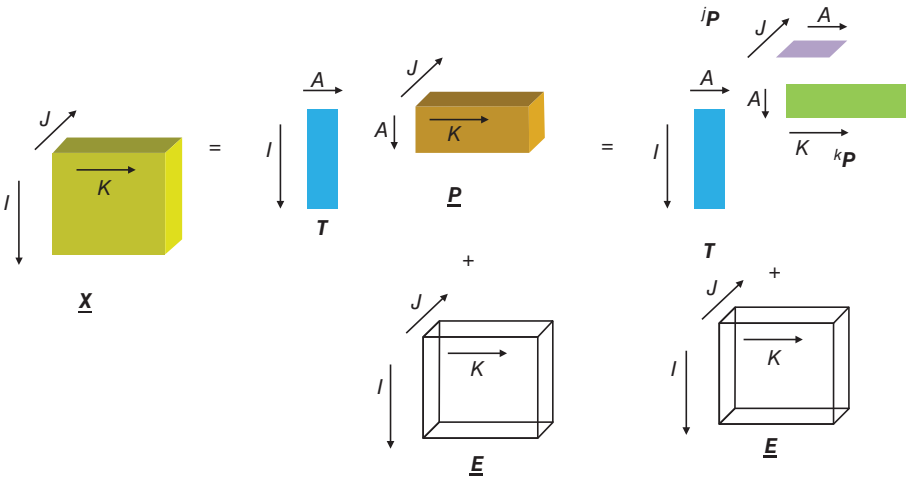


Figure 6.17 Representation of tri-linear PLS1.

the number of dimensions in both the x and c blocks; for example, a three-way x block and a two-way c block may consist of a series of two-way chromatograms, each containing several compounds. However, unless one has a good grip of the theory or there is a real need from the nature of the data, it is often more straightforward to reduce the problem to one of tri-linear PLS1: for example, a concentration matrix $\boldsymbol{C}$ can be treated as several concentration vectors, in the same way that a calibration problem that might appear to need PLS2 can be reduced to several calculations using PLS1.

Although there has been a huge interest in multi-mode calibration in the theoretical chemometrics literature, it is important to recognise that there are limitations to the applicability of such techniques. Good, very high order data are rare in chemistry. Even three-way calibration, such as in DAD–HPLC, has to be used cautiously, as there are frequent experimental difficulties with alignments of chromatograms in addition to interpretation of the numerical results. However, there have been some significant successes in areas such as sensory research and psychometrics and certain techniques such as fluorescence excitation emission spectroscopy, where the wavelengths are very stable, have demonstrated significant promise.

Using genuine three-way methods (even when redefined in terms of matrices) differs from unfolding in that the connection between different variables is retained; in an unfolded data matrix, there is no indication if two variables, for example, share the same elution time or spectroscopic wavelength for each joint variable such as a wavelength elution time pair (e.g. the absorbance at 300 nm and 20.2 min) is independent.

1. A $\boldsymbol{c}$ or concentration vector is estimated via PLS using $\hat{\boldsymbol{c}} = \boldsymbol{Tq}$.
   - **(a)** Using two-way and three-way PLS, the first element of $\boldsymbol{q}$ is the same, no matter how many components.
   - **(b)** Using two-way PLS, the first element of $\boldsymbol{q}$ is the same, no matter how many components, but differs according to the number of components for three-way PLS.
   - **(c)** Using two-way and three-way PLS, the first element of $\boldsymbol{q}$ differs according to the number of components.

6.6 Model Validation and Optimisation

Unquestionably, one of the most important aspects of all calibration methods is model validation. Numerous questions need to be answered.

- How many significant components are needed to characterise a data set to optimise the model?
- How well is an unknown predicted?
- How representative are the data used to produce a model?

Using auto-predictive or training set models, it is possible to obtain a fit as close as desired using more and more PLS or PCA components, until the raw data are fitted exactly; however, the later terms are unlikely to be physically meaningful and may model noise. There is a large literature on how to decide what model to adapt, which requires an appreciation of model validation, experimental design and how to measure errors. Most methods aim to guide the experimenter as to how many significant components to retain. The methods are illustrated below with reference to both PCR and PLS1, but similar principles apply to all calibration methods, including MLR, PLS2 and tri-linear PLS1. The most common question asked is how many components should be retained, and then once the optimum has been determined, how well an independent test set is predicted.

6.6.1 Auto-prediction

The simplest approach to determining number of significant components is by measuring the auto-prediction error. This is also called the root mean square error of calibration (*RMSEC*). Usually (but not exclusively), the error is calculated on the concentration data matrix ($\boldsymbol{c}$), and we will restrict the discussion below to errors in concentration for brevity: it is important to understand that similar equations can be obtained for the x block.

As more components are calculated, the residual error reduces. It is normally calculated by

$$RMSEC = \sqrt{\frac{\sum_{i=1}^{I} (c_i - \hat{c}_i)^2}{I - A - 1}}$$

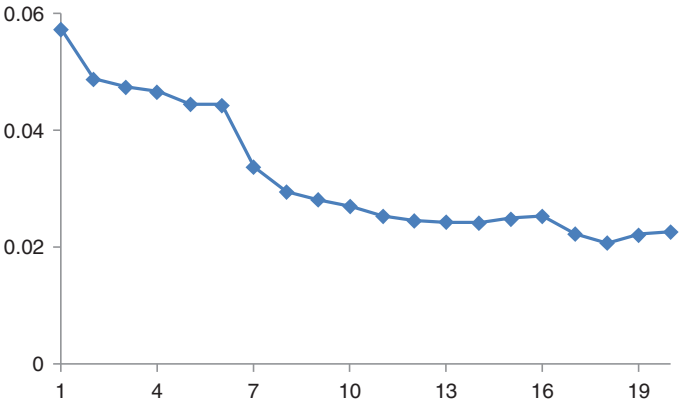


Figure 6.19 *RMSEC* auto-predictive errors for acenaphthylene using PLS1.

where A components have been used for the model and the data have been centred.

Note that these errors can easily be converted to a percentage variance or mean square errors as described in Sections 6.4 and 6.5. The value of *RMSEC* will usually decline in value as more components are calculated.

Note that there can sometimes be very slight increases, especially when the number of components is large, as although $\sum_{i=1}^{I}(c_i - \hat{c}_i)^2$ will never increase, the denominator will get smaller as the number of components increases.

The auto-predictive error for acenaphthylene is presented in Figure 6.19, using PLS1. A similar calculation can be done for PCR.

The auto-predictive error can be used to determine how many components to use in the model, in a number of ways.

- A standard cut-off percentage error can be used, for example, 1%. Once the error has reduced to this cut-off, ignore later PLS (or PCA) components.
- Sometimes, an independent measure of the noise level is possible. Once the error has declined to the noise level, ignore later PLS (or PCA) components.
- Occasionally, the error can reach a plateau. Take the PLS components up to this plateau.

By plotting the magnitudes of each successive component (or errors in modelling the x block), it is also possible to determine prediction errors for the x block. However, the main aim of calibration is to predict concentrations rather than spectra; thus, this information, although quite useful, is less frequently employed in calibration. More details have been discussed in the context of PCA in Section 4.3.2 and also Section 6.4.2, the ideas for PLS are similar.

Many statistically oriented chemometricians do not like to use auto-prediction determining the number of significant components, as it is always possible to fit data perfectly simply by increasing the number of terms (or components) in the model. There is, though, a difference between statistical and chemical thinking. A chemist might know (or have a good intuitive feel) for parameters such as noise levels and, therefore, in some circumstances, be able to successfully interpret the auto-predictive errors in a perfectly legitimate physically meaningful manner.

1. The root mean square error of calibration may not always decrease when more components are added to the model, using the equation presented in this section.
   (a) True
   (b) False

6.6.2 Cross-validation

An important chemometric tool is called *cross-validation*. The basis of the method is that the predictive ability of a model formed on part of a data set can be tested by how well it predicts the remainder of the data and has been introduced previously in other contexts (see Sections 4.3.2.2 and 5.5.2).

It is possible to determine a model using $I - 1$ (=24) samples leaving out one sample (i). How well does this model fit the original data? Below we describe the method when the data are centred, the most common approach.

For PCR, the following steps are used to determine cross-validation errors using leave one out (LOO).

has a minimum after nine components are included in the model and then increases steadily afterwards. By inspecting the minimum, we can estimate that there are nine significant PLS components in the model. There are no hard and fast rules though, and it is important not to be carried away by automated approaches. Note that a different number of significant components may be found for each compound in the mixture if using PLS1. In addition, different conclusions might be drawn from the *x* block.

Cross-validation has two main purposes.

- It can be employed as a method for determining how many significant components characterise the data. From Figure 6.20, it appears that nine components are necessary to obtain an optimal model for acenaphthylene. This number will not always equal the number of compounds in the mixture, as spectral similarity and correlations between concentrations will often reduce this, whereas impurities or interferents may increase it; our design has been carefully chosen so that each compound's concentration profile is orthogonal, but in most real-world situations, this will not be so. Later components probably model noise and it would be unwise to include them if using to predict concentrations of unknowns.
- It can be employed as an estimate of predictive ability. The minimum cross-validated prediction error for acenaphthylene of 0.0493 mg/l equals 41.1% of the mean. This compares with an auto-predictive error of 0.0281 mg/l or 23.44% using nine components and PLS1, which is an over-optimistic estimate. The cross-validated error is almost always higher than the auto-predictive (or calibration) error and provides an indicative of what the error would be if we were to estimate the concentration of an unknown sample.

Many refinements to cross-validation have been proposed in the literature, which have been discussed in Chapter 4; it is equally possible to apply these to calibration as well as pattern recognition.

1. Calculating the *RMSEC* and *RMSECV* for an eight-component model using centred data and 40 samples involves
   (a) 40 degrees of freedom for *RMSEC* and 40 for *RMSECV*.
   (b) 31 degrees of freedom for *RMSEC* and 31 for *RMSEC*.
   (c) 31 degrees of freedom for *RMSEC* and 40 for *RMSECV*.
   (d) 40 degrees of freedom for *RMSEC* and 31 for *RMSECV*.

6.6.3 Independent Test Sets

A significant weakness of cross-validation is that the results depend on the design and scope of the original data set used to form the model. This data set is often called a *training set* (see Chapter 5), although some authors prefer the term *calibration set*. Consider a situation in which a series of mixture spectra are recorded, but it happens that the concentration of two compounds are correlated, so that the concentration of compound A is high when compound B likewise is high, and *vice versa*. A calibration model can be obtained from analytical data, which predicts both concentrations quite well. Even cross-validation might suggest the model is good. However, if asked to predict a spectrum where compound A is in a high concentration and compound B in a low concentration, it is likely to give very poor results, as it has not been trained to cope with this new situation. Cross-validation is very useful for removing the influence of internal factors such as instrumental noise or dilution errors or instrumental noise but cannot help very much if there are correlations in the concentration of compounds in the training set. If new data have a different correlation structure to the training (or calibration) set, predictions may be very poor. This is a huge dilemma in chemometrics: we cannot predict the composition of all possible future samples, so should the training set be very broad in scope, in which case the model may be quite poor or the sample size has to be very large, or should it be quite narrow and so predicted effectively and on a modest number of samples but not robust to future samples with a different structure. There is no obvious answer.

In some cases, there will be inevitable correlations in the concentration data because it is not easy to find samples without this. Examples routinely occur in environmental monitoring. Several compounds often arise from a single source. For example, PAHs are well-known pollutants; hence, if one or two PAHs are present in high concentrations, it is a fair bet that others will be too. There may be some correlations, for example, in the occurrence of compounds of different molecular weights if a homologous series occurs, for example, as the by-product of a biological pathway, there may be an optimum chain length, which is most abundant in samples from a certain source. It would be hard to find a series of field samples in which there are no correlations between the concentrations of compounds. Consider, for example, setting up a model of PAHs coming from rivers close to several specific sources of pollution. The model may